

Task 1: Instagram Data Analysis Report

1. Executive Summary

This report presents an Instagram engagement analysis completed as part of the Data Science Internship at InternSpark. The main objective was to understand how different factors such as posting time, posting day, content type, filter usage, hashtags, likes, comments, and follower count influence engagement.

The dataset contained multiple Instagram-related CSV files, including users, photos, likes, comments, follows, tags, and photo-tag mappings. These files were cleaned, standardized, and merged to create a post-level analysis table. Key engagement metrics such as likes count, comments count, total engagement, follower count, likes per follower, comments per follower, and engagement rate were calculated.

The analysis found that posts around 8:00 AM and on Thursday showed the highest average engagement in the available dataset. Photo-based posts performed slightly better than other content types, while posts without Instagram filters showed a small engagement advantage. Hashtag analysis showed that #delicious had the highest average engagement among the analyzed hashtags.

Based on these insights, a practical Instagram growth strategy was prepared for Alfido Tech, focusing on consistent posting, photo-based content, relevant hashtags, and regular engagement tracking.

2. Objective

The objective of this analysis is to study Instagram engagement data and identify patterns related to:

1. Best posting time and day
2. Content type performance
3. Filter usage impact
4. Hashtag usage and hashtag performance
5. Likes, comments, follower count, and engagement rate
6. Practical content strategy recommendations for Alfido Tech

3. Dataset Overview

The dataset consisted of multiple Instagram-related CSV files. Instead of working with one single table, the analysis required combining multiple connected files.

File Name	Description
users.csv	Contains user details such as user ID, account type, post count, and verification status
photos.csv	Contains post/photo details such as photo ID, user ID, created date, filter usage, and photo type
likes.csv	Contains like activity for each photo
comments.csv	Contains comment activity for each photo
follows.csv	Contains follower-followee relationship data
tags.csv	Contains hashtag/tag details
photo_tags.csv	Connects photos with hashtags

The uploaded dataset output confirmed that the files include data such as comments, follows, likes, photo tags, photos, tags, and users.

4. Tools and Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Google Colab
- GitHub

5. Data Preparation

The raw dataset had inconsistent column names such as extra spaces, mixed capitalization, and different naming formats.

To prepare the data for analysis:

- Column names were cleaned and standardized.
- Date and time fields were converted into proper datetime format.
- Posting hour and posting day were extracted from the post timestamp.
- Likes and comments were grouped by photo ID.
- Follower count was calculated using follow relationship data.
- Multiple files were merged to create one post-level analysis table.

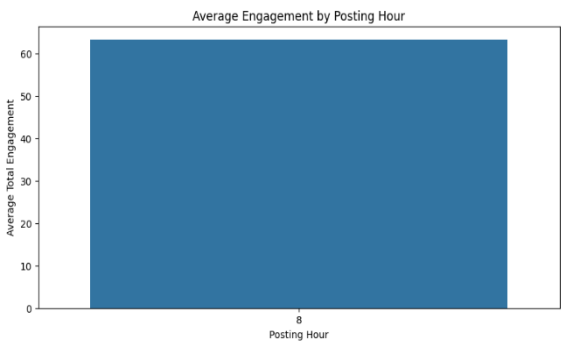
6. Metrics Calculated

The following engagement metrics were created:

Metric	Formula / Meaning
Likes Count	Total number of likes received by a post
Comments Count	Total number of comments received by a post
Total Engagement	Likes Count + Comments Count
Follower Count	Number of followers for the post owner
Likes per Follower	Likes Count / Follower Count
Comments per Follower	Comments Count / Follower Count
Engagement Rate	Total Engagement / Follower Count

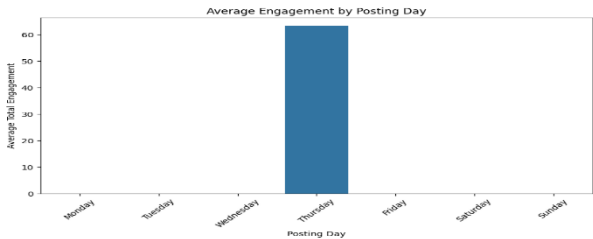
7. Key Visualizations

7.1 Average Engagement by Posting Hour



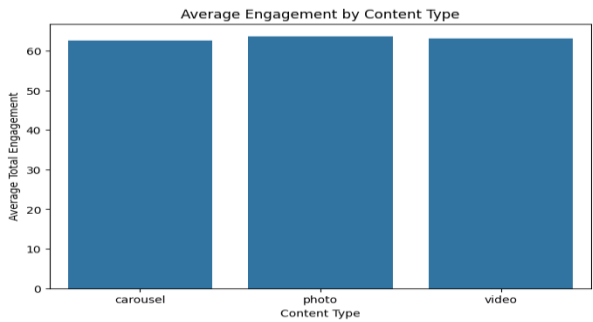
This chart was used to identify which posting hour received the highest average engagement.

7.2 Average Engagement by Posting Day



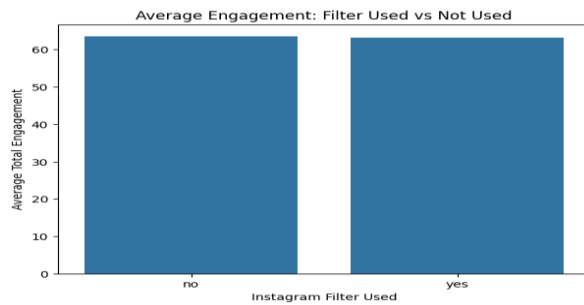
This chart helped compare engagement across different days of the week.

7.3 Engagement by Content Type



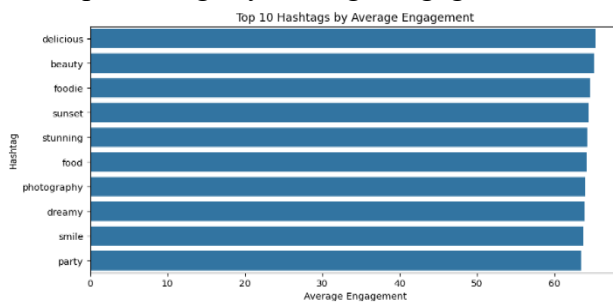
This chart compared average engagement across available content types.

7.4 Filter Usage vs Engagement



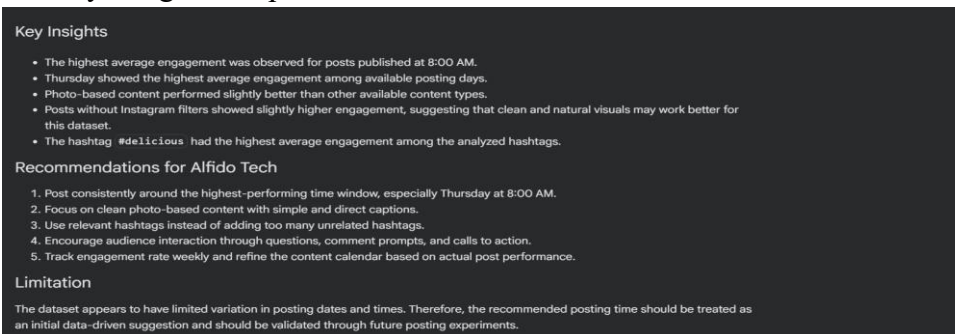
This chart analyzed whether posts with Instagram filters or without filters performed better.

7.5 Top Hashtags by Average Engagement



This chart helped identify hashtags associated with stronger average engagement.

7.6 Key Insights Output



8. Key Findings

Analysis Area Finding

Best Posting Hour	8:00 AM showed the highest average engagement
Best Posting Day	Thursday showed the highest average engagement
Best Content Type	Photo-based content performed slightly better
Filter Usage	Posts without filters showed slightly higher engagement
Best Hashtag	#delicious had the highest average engagement
Average Engagement at Best Time	63.31
Average Engagement for Photo Content	63.63
Average Engagement Without Filter	63.50
Average Engagement for #delicious	65.27

9. Strategy Recommendations for Alfido Tech

Based on the analysis, Alfido Tech can follow these content strategy recommendations:

Post during high-engagement windows

Based on the available dataset, Thursday around 8:00 AM showed the strongest engagement.

Focus on clean photo-based content

Photo-based posts performed slightly better, so Alfido Tech can use clear visuals, project snapshots, learning updates, and student-focused content.

Use relevant hashtags, not excessive hashtags

Hashtags should match the content theme. Random hashtag stuffing should be avoided.

Encourage audience interaction

Captions should include questions, calls to action, or prompts that encourage comments.

Track engagement weekly

Alfido Tech should monitor likes, comments, follower count, and engagement rate regularly to improve future content decisions.

10. Recommended Weekly Content Calendar

Day	Recommended Content	Goal
Monday	Educational post or tech/data tip	Awareness
Tuesday	Behind-the-scenes or student learning content	Trust building
Wednesday	Project showcase or internship update	Portfolio visibility
Thursday	High-priority photo post around 8:00 AM	Maximum engagement
Friday	Question, poll, or interaction post	Comments and engagement
Saturday	Community or student achievement post	Community connection
Sunday	Weekly recap or learning summary	Consistency

11. Limitation

The dataset appears to have limited variation in posting dates and times. Therefore, the recommendation that Thursday at 8:00 AM performs best should be treated as an initial data-driven suggestion, not a universal conclusion. This strategy should be validated further using future Instagram post data.

12. GitHub / Notebook Link

GitHub Task Folder:

<https://github.com/saitharun-student-27/InternSpark-Data-Science-Internship/tree/main/Task-1-Instagram-Data-Analysis>

13. Conclusion

This task helped convert raw Instagram data into meaningful engagement insights. By cleaning multiple CSV files, combining related tables, calculating engagement metrics, and creating visualizations, the analysis produced useful recommendations for Alfido Tech's Instagram content strategy.

The main learning from this task was that data analysis is not only about writing code or creating charts. It is about preparing messy data, asking the right questions, identifying useful patterns, and converting those patterns into practical decisions.